ds9-rust

User Manual — v0.1

A Rust + slint port of SAOImage DS9

A pragmatic, OGFinder-style FITS
viewer for working astronomers —
multi-frame, region-aware, WCS-correct, scriptable.

Smithsonian Astrophysical Observatory's SAOImage DS9, reimagined in Rust.
github.com/kjhan0606/rusty_ds9

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Contents

1 Quick start

```
# build (release recommended for large mosaics + smooth / bin)
cargo build --release -p ds9-app

# launch — open files via the File menu
./target/release/ds9

# preload an image (with optional region file + catalog)
./target/release/ds9 deep_field.fits regions.reg sources.cat
```

Note

The binary is named `ds9`, not `ds9-app`. The crate name and binary name diverge on purpose — it lets you alias `ds9` on your `PATH` without colliding with the upstream Tcl/Tk DS9 if you keep both installed.

System requirements. Rust $\geq$ 1.85, a working OpenGL stack (slint defaults to a winit + femtovg backend), and an X11 / Wayland / macOS Cocoa display. Optional: `source-extractor` (or `sextractor`, or `sex`) on `PATH` for the catalog wrapper; `lpr` for printing.

1.1 Supported inputs

Image cube	`.fits`, `.fit`, `.fts`, `.fz` (Rice tile-compressed), NAXIS=2 or NAXIS=3
Region file	DS9 `.reg` — circle / box / ellipse / annulus / point / line / polygon, in `image` or WCS coordinates (`fk5` / `icrs` / `galactic`)
Catalog	TSV, whitespace, CSV, SExtractor ASCII_HEAD, minimal VOTable (`<TR>/<TD>`)
Colormap	256-stop user LUT (text `R G B` per row, 0–1 or 0–255)

2 Core concepts

Frames. Every loaded image is a *frame*. A frame owns its own stretch, limits, colormap, regions, catalog, view zoom & pan, smoothing, binning, orientation, and (for cubes) 3-D projection mode. The info bar's frm cell shows `<active>/<total>`.

The render pipeline runs in this order, every refresh:

bin $\rightarrow$ smooth $\rightarrow$ stretch + limits $\rightarrow$ colormap (LUT-aware) $\rightarrow$ orientation

Coordinates. Every mouse handler converts display (x, y) to FITS pixel $(x_{\text{fits}}, y_{\text{fits}})$ via `display_to_fits()` so flips, 180° rotations, and crops never leak into region geometry. WCS readouts use a TAN gnomonic projection parsed from `CRPIX/CRVAL/CD_*`.

3 The window

Menubar	File / Edit / View / Frame / Bin / Zoom / Scale / Color / Region / Catalog / Analysis / Help.
Mode toggles	`pan` / `edit` / `region` / `crosshair` (next to the info bar).
Info bar	cursor x/y, pixel value, WCS readout, frame count, mode.
Sidebar	catalog table, marker list, optional panner / magnifier.
Canvas	the image. Drag to pan; click to interact per mode.
Colorbar	vertical colorbar with the active stretch + limits.

3.1 Modes

pan drag = pan; click on a region selects it.
edit drag = pan or move a region; click empty canvas to drop a small circle marker.
region click drops the active region template (default circle); the property editor (Region ▷ Property) lets you tune the just-placed region.
crosshair click pins a session crosshair in WCS (or pixel) space. The crosshair follows the active frame: switch frames and the same on-sky position is re-projected.

4 Menubar reference

The menubar is the canonical surface — every feature is reachable without an IPC client.

4.1 File

Open…	file picker; loads into a new frame.
Save Image…	PNG of the current canvas (after stretch/cmap).
Save FITS…	BITPIX=-32 float export with WCS preserved.
Save TIFF…	in-house TIFF writer (no libtiff dep).
Save EPS…	PostScript dump of the colour-mapped raster.
Print…	renders to a temp PNG then `lpr`'s it.
SAMP Send Image	broadcasts the current FITS path over SAMP (see §15).
SAMP Send VOTable	broadcasts a saved VOTable to other SAMP clients.
Quit	exit.

4.2 Edit

Undo / Redo	not yet — noop placeholders.
Crop	sets the visible region of the active frame (zoom + pan adjusted; original FITS untouched). *Reset Crop* restores fit.
Preferences…	toggles the Preferences panel (cmap / stretch / limits / smooth / bin / blink-ms). `Apply` pushes onto the active frame; `Save` persists to `~/.config/ds9-rust/prefs.toml`.

4.3 View

Toggles the panner, magnifier, coordinate-grid overlay, crosshair, info panel, mode buttons, and colorbar. All toggles persist for the session.

4.4 Frame

New Frame / Delete Frame	opens picker / removes active.
Next / Previous	wraps.
Match…	broadcast active frame's view (zoom / pan / cmap / scale — per matching lock) to siblings.
Blink	cycles every 500 ms.
RGB Composite	frames 1, 2, 3 → R, G, B channels of a new frame.
HDU Movie	auto-cycles HDUs of the active file every 800 ms.
Mosaic WCS	WCS-aware reprojection of all open frames into one (§9.1).
Tile Frames	toggles a grid view of all open frames; click a tile to make it active.
3D Slice…	for NAXIS=3 cubes: cycles through planes one click at a time.
3D Max Intensity	projects max along the cube axis.
3D Sum	sum projection.
3D Mean	mean projection (NaN-safe).
Rotate 180° / Flip H / Flip V / Reset Orientation	display-time transforms; coords stay native.
Lock Zoom / Pan / Color / Scale	scope of `Match…` and `Blink`.

4.5 Bin

1 / 2 / 4 / 8 / 16 / 32	block-bin factor (active frame).
Average / Sum / Sub-sample	reduction mode for the bin operator.

4.6 Zoom

In, Out, Fit (canvas), Reset (1:1), or pick a fixed $1\times/2\times/4\times/8\times/16\times$.

4.7 Scale

Stretches: `linear`, `log`, `sqrt`, `squared`, `asinh`, `sinh`.
Limits: `zscale`, `minmax`.

4.8 Color

Built-in colormaps: `grey, red, green, blue, a, b, bb, heat, cool, rainbow, sls, hsv`.
Load Custom… reads a 256-row LUT (`R G B` text, 0–1 or 0–255); *Clear Custom* drops back to the built-in.

4.9 Region

New / Load…/ Save…/ Delete Selected / Delete All / Info / Property…(opens the editor for the currently selected region — edit centre, radius, rotation, colour, label, then Apply).

4.10 Catalog

Load…, Clear, Run SExtractor…(§8.1), Online Query…(§8.2), Info.

4.11 Analysis

Pixel Table…	11×11 raw values around the cursor.
Statistics…	frame-wide N, min, max, mean, median, σ, and finite-pixel count.
Histogram…	128-bin log-y histogram, computed against the current stretch limits.
Contour Levels…	5-level overlay (sign-change between zscale low/high).
Smooth (cycle)	cycles Gaussian $\sigma \in \{0, 2, 4, 8\}$ px.
Smooth Off	disables.
Smooth Kind…	switches between Gaussian, Boxcar, Median kernels.
Centroid	flux-weighted (x, y) + half-light radius for the selected region.
Radial Profile	1-D azimuthal profile around the selected region; opens a floating plot.
Projection	line-pixel profile for a *line* region.

5 Mouse & keyboard

Pan	drag (pan mode); arrow keys nudge by 1 canvas pixel.
Zoom	menu, scroll-wheel, or IPC `zoom in/out/N`.
Hover readout	x/y, value, WCS in the info bar.
Select region	click on it (any mode).
Move region	edit-mode press-and-drag.
Drop region	edit-mode click on empty canvas (default circle, $r = 6$).
Pick catalog source	click near the symbol or click the row in the table.
Crosshair	crosshair-mode click (WCS-pinned if frame has WCS).

6 Working with regions

DS9 region files are read and written in their canonical form. Image-coord regions look like:

```
# Region file format: DS9 version 4.1
image
circle(512.0, 256.5, 18.0) # color=cyan text={hot pixel}
box(120, 80, 40, 25, 0)
ellipse(720, 400, 50, 30, 35) # color=yellow
annulus(960, 540, 8, 20)
point(160, 160) # point=cross
line(50, 50, 200, 200) # color=magenta
polygon(100, 100, 200, 110, 180, 200, 90, 180)
```

Properties carried. `color=`, `width=`, `text={...}`, dash, font. The Region ▷ Property dialog edits all of these for the selected region.

7 WCS-aware regions

When the active frame has a WCS, DS9-style WCS regions load correctly. The following all parse:

```
fk5
circle(12:34:56.7, +25:18:12, 30")
ellipse(189.123, 45.678, 1.5', 0.8', 30)
box(10h12m05s, -27d05'13", 2', 1', 0)
polygon(189.1, 45.7, 189.15, 45.72, 189.13, 45.78)
```

Sizes. " = arcsec, ' = arcmin, d = deg. Decimal-degree numbers without a suffix are also accepted.

No WCS? The parser falls back to `image` semantics and the status bar warns you so you can spot the silent reinterpretation.

8 Source catalogs

Auto-detect. The loader sniffs the first ∼256 bytes:

- `<VOTABLE>` or `<TABLE>` → VOTable parser.
- Leading # on first non-blank line → SExtractor.
- Tab in the first row → TSV (header row).
- Comma in the first row → CSV (header row, quoted strings OK).
- Otherwise → whitespace-separated, columns synthesised (`col1, col2, …`).

X / Y mapping. The catalog overlay needs image coords. The loader looks up `X_IMAGE/Y_IMAGE` (SExtractor), falling back to `XWIN_IMAGE/YWIN_IMAGE`, then bare `X/Y` or `x/y`.

8.1 Run SExtractor

Catalog ▷ Run SExtractor… shells out to the external SExtractor binary on the active frame's source FITS, parses `ASCII_HEAD` output, and overlays the result. Defaults: `DETECT_THRESH=1.5`, `DETECT_MINAREA=5`, `BACK_SIZE=64`. Override per-key via `SEXTRACTOR_OPTS`:

```
SEXTRACTOR_OPTS="-DETECT_THRESH 3.0 -DEBLEND_MINCONT 0.0001" \
  ./target/release/ds9 image.fits
```

The IPC verb `sextractor` runs the same code path.

8.2 Online query

Catalog ▷ Online Query… opens a small panel with three buttons:

Resolve	SIMBAD/Sesame name → (RA, Dec). Recenters the active frame on the resolved position; drops a crosshair.
VizieR	cone search around the cursor / crosshair (radius in arcmin). Result loads as a VOTable catalog.
NED	NED `objsearch` cone search (returns a bar-separated ASCII table).

All requests use HTTPS via `ureq + rustls` (no system OpenSSL).

9 Multi-frame, Match, Blink, RGB

Every File ▷ Open… pushes a new frame. Frame ▷ Next / Previous cycles them; per-frame view state (zoom, pan, stretch, limits, cmap, regions, catalog, smooth, bin) is saved on switch and restored when you come back.

Match. Broadcasts the active frame's state to siblings. The Lock Zoom / Pan / Color / Scale toggles control which channels are pushed. `Lock Color` pushes both built-in cmaps and any custom LUT.

Blink. Cycles all frames every 500 ms. Combine with `Lock Pan` for true side-by-side comparison.

RGB Composite. Frame ▷ RGB Composite takes frames 1–3 (whichever are loaded), zscale-stretches each, packs them into the R/G/B channels of a new frame, and pushes it onto the stack. Useful for tri-band difference imaging.

9.1 Mosaic WCS

Frame ▷ Mosaic WCS reprojects every open frame that has a WCS into a single output frame. The output WCS is the first WCS-bearing frame's, shifted to span the bounding box of all corners. The reprojector uses output-driven nearest-neighbour: world$(out) \to$ pix(in) for each output pixel. Overlap is averaged. Frames without a WCS are skipped (with a warning).

9.2 Tile Frames

Toggles a grid view of all open frames (cols $= \lceil \sqrt{n} \rceil$). Each tile is a per-frame RGBA composite scaled with aspect preservation. Click any tile to switch to that frame and exit tile mode.

10 Filters: smoothing & binning

Smoothing. Gaussian, Boxcar, or Median kernel; per-frame. Analysis ▷ Smooth (cycle) steps Gaussian σ through $0 \to 2 \to 4 \to 8$ px. Smooth Off resets. Smooth Kind… switches kernel.

Binning. Bin ▷ N block-bins the displayed image $N \times N$. Bin ▷ Average / Sum / Sub-sample picks the reduction. Visualisation only — the underlying FITS pixel grid and WCS are untouched.

11 3-D cube rendering

When the active frame loaded a NAXIS=3 cube, the full plane stack is held on the side; the 2-D display starts on plane 1. Use the Frame menu's 3D entries to project:

3D Slice…	each click advances to the next plane (with wrap). IPC `3d slice N` sets a specific 1-based index.
3D Max Intensity	per-pixel max along the cube axis (NaN-safe).
3D Sum	per-pixel sum (skips NaN).
3D Mean	per-pixel mean (skips NaN, divisor is finite count).

The chosen mode is sticky on the frame — smoothing, binning, stretch, and cmap all run downstream of the projection.

12 Analysis tools

12.1 Statistics

Frame-wide finite count, min / max, mean, median, σ.

12.2 Histogram

128-bin, log-y, computed against the current stretch limits.

12.3 Contour overlay

5 levels via sign-change detection between zscale low/high. Re-rendered on every refresh; orientation-aware.

12.4 Pixel table

11×11 raw pixel values centred on the cursor.

12.5 Centroid

Flux-weighted centroid plus a rough half-light radius for the selected region (typically a circle / box / ellipse).

12.6 Radial profile

Azimuthally averaged 1-D profile, drawn in a floating plot window.

12.7 Projection

For a *line* region, samples the pixel values along the line and plots them. Useful for slit-style profile checks.

12.8 Crop

Edit ▷ Crop sets the active frame's visible region to the current view. Reset Crop returns to a full-frame fit. The underlying FITS image is never modified.

13 Saving, exporting, printing

Save Image…	PNG of the current canvas (RGBA, alpha untouched).
Save FITS…	BITPIX=−32, WCS keywords preserved, BSCALE=1 / BZERO=0.
Save TIFF…	in-house writer; no libtiff.
Save EPS…	PostScript dump of the colour-mapped raster.
Save JPEG…	not built in (returns a status message).
Print…	renders to a temp PNG, then pipes via `lpr`.

14 IPC protocol

ds9-rust ships a line-based, XPA-flavoured IPC protocol. Two transports listen by default:

- Unix-domain socket at `$XDG_RUNTIME_DIR/ds9-rust-$USER.sock` (or `/tmp/…` on macOS).
- TCP loopback on a dynamic port (e.g. `127.0.0.1:54861`).

The chosen paths/ports are written to `~/.ds9-rust/xpa.info` so external clients can discover them:

```
pid=12345
unix=/run/user/1000/ds9-rust-alice.sock
tcp=54861
```

14.1 Verb grammar

quit, exit	terminate.
xpaaccess, version	XPA-style probes.
mode M	set window mode (pan / edit / region / crosshair).
frame N, frame next/prev	switch frame.
frame new / delete	menu equivalents.
frame mosaic / tile	menu equivalents.
scale linear|log|sqrt|...	set stretch.
scale limits LO HI	set user limits.
cmap NAME	switch built-in colormap.
bin N	block-bin factor.
zoom in|out|fit|reset|N	view zoom.
pan to X Y	pan in image coords.
region load / save PATH	DS9 .reg I/O.
file open PATH	load a FITS file (new frame).
save png|fits|tiff|eps P	export.
value	emit the pixel value at the current cursor.
sextractor	run the SExtractor wrapper.
samp image / votable PATH	broadcast over SAMP.
hdu next / list / N	HDU navigator.
movie on / off	toggle HDU movie.
3d max / sum / mean	cube projection.
3d slice N	cube slice index (1-based).
3d depth	query cube depth.
help	one-line summary.

14.2 Examples

```
SOCK=$(awk -F= '/^unix=/{print $2}' ~/.ds9-rust/xpa.info)

echo "frame next"             | nc -U "$SOCK"
echo "scale log"              | nc -U "$SOCK"
echo "cmap heat"              | nc -U "$SOCK"
echo "region load /tmp/x.reg" | nc -U "$SOCK"
echo "save png /tmp/out.png"  | nc -U "$SOCK"
echo "3d max"                 | nc -U "$SOCK"
echo "3d slice 12"            | nc -U "$SOCK"
echo "value"                  | nc -U "$SOCK"
echo "help"                   | nc -U "$SOCK"
```

The TCP transport accepts the same grammar:

```
PORT=$(awk -F= '/^tcp=/{print $2}' ~/.ds9-rust/xpa.info)
echo "version" | nc 127.0.0.1 "$PORT"
```

Heads up

TCP listens only on `127.0.0.1`. Don't expose ds9-rust on a public interface — there's no auth on the IPC channel.

15 SAMP messaging

File ▷ SAMP Send Image / VOTable broadcasts the current FITS path (or a saved VOTable) to other SAMP clients running on the same desktop — TOPCAT, Aladin, etc.

How it works. ds9-rust reads the standard SAMP lockfile at `~/.samp`, registers a client with the hub via XML-RPC, then sends `samp.hub.notifyAll` with the canonical `image.load.fits` or `table.load.votable` mtype. The XML-RPC serialiser is hand-rolled (no SOAP / glib dep). On exit the client unregisters cleanly.

Caveats. Outbound only; ds9-rust does not currently subscribe to incoming SAMP messages.

16 Preferences

Edit ▷ Preferences… toggles a panel. Editable: default colormap / stretch / limits / smoothing string / bin factor / blink interval (ms). Buttons:

- **Apply** — push onto the active frame.
- **Save** — persist to `~/.config/ds9-rust/prefs.toml` (a simple `key = value` text file, not real TOML).
- **Reset** — restore built-in defaults.

The prefs file is loaded automatically at startup; new frames inherit those defaults.

17 Cookbook

Open a deep-field stack and bring up SExtractor sources

```
./target/release/ds9 deep_field.fits
# Catalog -> Run SExtractor...   (uses defaults; tweak SEXTRACTOR_OPTS if needed)
# Click a star on the image -> table jumps; click a row -> view recenters.
```

RGB difference image from three exposures

```
SOCK=$(awk -F= '/^unix=/{print $2}' ~/.ds9-rust/xpa.info)
echo "file open r.fits" | nc -U "$SOCK"
echo "file open g.fits" | nc -U "$SOCK"
echo "file open b.fits" | nc -U "$SOCK"
echo "frame 1; cmap red"   | nc -U "$SOCK"
echo "frame 2; cmap green" | nc -U "$SOCK"
echo "frame 3; cmap blue"  | nc -U "$SOCK"
# Frame -> RGB Composite  (creates a new packed frame)
```

Walk through a spectral cube

```
./target/release/ds9 cube.fits
# Frame -> 3D Slice...   (each click advances)
# Or via IPC:
echo "3d slice 17" | nc -U "$SOCK"
echo "3d max"      | nc -U "$SOCK"   # then back to a max-intensity view
```

Mosaic four CCD chips into one frame

```
./target/release/ds9 chip1.fits chip2.fits chip3.fits chip4.fits
# Frame -> Mosaic WCS
# (output frame uses chip1's WCS, shifted to fit all chip corners)
```

18 Troubleshooting

No matching fbConfigs / GLXBadFBConfig on launch	slint's OpenGL backend can't find a config. Try `SLINT_BACKEND=software ./target/release/ds9`, or update Mesa / your GPU driver.
RefCell already borrowed when opening a file dialog (macOS)	fixed in 4fa677f. If you're on an older build, pull `main`.
file contains no 2-D image HDU	The file is a pure binary table or its primary HDU has `NAXIS=0`. Use Frame ▹ HDU Movie or the HDU navigator (sidebar) to step through extensions.
tile-compressed .fz precision loss	Make sure the workspace `[patch.crates-io]` for vendored `fitsrs` is intact. The patch upgrades unquantize to f64.
SAMP Send Image silently does nothing	No SAMP hub running; start TOPCAT or Aladin first.

19 Architecture

ds9-app	slint UI shell; all event wiring; menubar handler; frame / state management; IPC dispatch; mosaic + tile + 3D + SAMP + online-query glue.
ds9-fits	FITS I/O wrapping vendored `fitsrs`; BITPIX 8/16/32/−32/−64, BSCALE/BZERO/BLANK; tile-compressed `.fz`; minimal TAN-projection WCS; NAXIS=3 cube loader.
ds9-image	stretches (linear/log/sqrt/squared/asinh/sinh), limits (zscale/minmax), built-in + custom colormaps, RGBA renderer, smoothing kernels, block-bin operators.
ds9-marker	DS9 `.reg` parser/writer + region/marker model (image-coord shapes, with WCS round-trip handled by ds9-app).
ds9-catalog	TSV / SExtractor / CSV / VOTable parsers, column lookup, numeric sort.
vendor/fitsrs	local `fitsrs` patch (f64 unquantize for `.fz`).

Persistent files written by ds9-rust:

- `~/.config/ds9-rust/prefs.toml` — preferences.
- `~/.ds9-rust/xpa.info` — IPC discovery (pid, unix socket path, tcp port).